Name: Pankaj Kumar Roll No. 230733

Laboratory Report – Exercise 9

Data Collection using Terrestrial Laser Scanning (TLS)

1. Objective

The primary objective of this laboratory exercise is to gain hands-on experience in collecting three-dimensional spatial data using a Terrestrial Laser Scanner (TLS). Students will learn to operate the Riegl TLS instrument, configure appropriate scanning parameters, place registration targets strategically, execute scans from multiple positions, and perform basic post-processing using Riegl RiScan Pro software. The exercise also aims to develop an understanding of point cloud quality assessment, including analysis of data density, intensity variation, and shadowing effects.

2. Instruments and Equipment

The following instruments and materials were used during the laboratory session:

- Riegl TLS (Terrestrial Laser Scanner)
- Tripod for stable scanner mounting
- Registration targets (spherical / flat reflective targets)
- Camera mounted on TLS for RGB colorization of point clouds
- Measuring tape for ground-truth measurements
- Riegl RiScan Pro software (for data import, processing, and export)



3. Theory and Background

Terrestrial Laser Scanning (TLS) is a ground-based active remote sensing technology that captures dense, three-dimensional point clouds of surfaces using pulsed or phase-based laser ranging. The scanner emits laser pulses and records the time of flight (ToF) or phase difference of the returned signal to compute the distance to each target point. By combining range measurements with precise angular positioning, the scanner builds a three-dimensional representation of the environment.

Key concepts relevant to this exercise include:

- Point Cloud: A set of data points in 3D space representing the external surface of objects.
- Scan Registration: The process of aligning multiple scans from different positions into a single coordinate system using common reference targets.
- Return Intensity: The measure of laser energy reflected back to the scanner, dependent on material reflectivity and angle of incidence.
- Occlusion / Shadow Effect: Areas not visible from the scanner position, creating data gaps in the point cloud.
- Data Density: The number of points per unit area; influenced by scanner resolution settings, range, and scan speed.

4. Site Reconnaissance and Scan Planning

Before initiating the scanning process, a thorough reconnaissance of the scan area was conducted to determine optimal scanner positions and target placements. The following considerations guided the planning:

4.1 Scanner Position Selection

- Positions were chosen to maximize coverage of the object/scene of interest.
- Overlapping scan regions of at least 30% were ensured between adjacent scans to facilitate accurate registration.
- At least two scan positions were selected as required by the exercise protocol.
- Positions with stable, level ground were preferred to ensure the tripod remained steady throughout scanning.

4.2 Target Placement Strategy

- A minimum of three registration targets were placed within the overlapping regions visible from adjacent scan positions.
- Targets were positioned at varying heights and distances to avoid co-planar configurations, ensuring robust registration.
- Target positions were secured to prevent any movement during scanning sessions.
- The height of each target above the ground was measured with a measuring tape and recorded for use in the next exercise.



5. Scanner Configuration and Parameters

The following scanner parameters were configured in RiScan Pro prior to initiating each scan:

Parameter	Scan Position 1	Scan Position 2
Resolution / Angular Step	0.04°	0.04°
Scan Speed	300,000 pts/sec	300,000 pts/sec
Field of View (H)	360°	360°
Field of View (V)	100°	100°
Intensity Calibration	Enabled	Enabled
No. of Points Collected	~25–40 million	~25–40 million

6. Scanning Procedure

6.1 Setup and Scanner Levelling

- The TLS scanner was mounted on a sturdy tripod and levelled using the built-in electronic bubble level.
- The scanner was powered on and connected to the field laptop running RiScan Pro software.
- A new project was created in RiScan Pro and the scanner was initialized.

6.2 Target Detection and Scan Execution

- A preview/coarse scan was performed to identify target positions within the scene.
- Registration targets were detected and their reflectance signatures confirmed in the software.
- Fine scan parameters were set and the full scanning sequence was executed.
- Data completeness was verified on screen before moving the scanner to the next position.

6.3 Multiple Scan Positions

- The scanner was repositioned to the second predetermined location.
- The tripod was relevelled and the scanning process repeated.
- Common targets visible in both scans were confirmed before ending the field session.

8. Post-Processing in RiScan Pro

8.1 Data Import and Preprocessing

- Scans were imported into a new RiScan Pro project.
- Noise filtering was applied to remove erroneous returns (e.g., flying points, atmospheric reflections).
- Unwanted background points were removed using clipping/cropping tools.

8.2 Scan Registration

- Registration targets were identified and labelled in each scan.
- Coarse registration was performed by matching corresponding targets across scans.
- Fine registration (ICP – Iterative Closest Point) was applied to minimise residual errors.
- Registration accuracy (RMS error) was noted and assessed.

8.3 Point Cloud Colorization

- The mounted camera images were used to assign RGB colour values to the point cloud.
- Colour calibration was verified and applied in RiScan Pro.

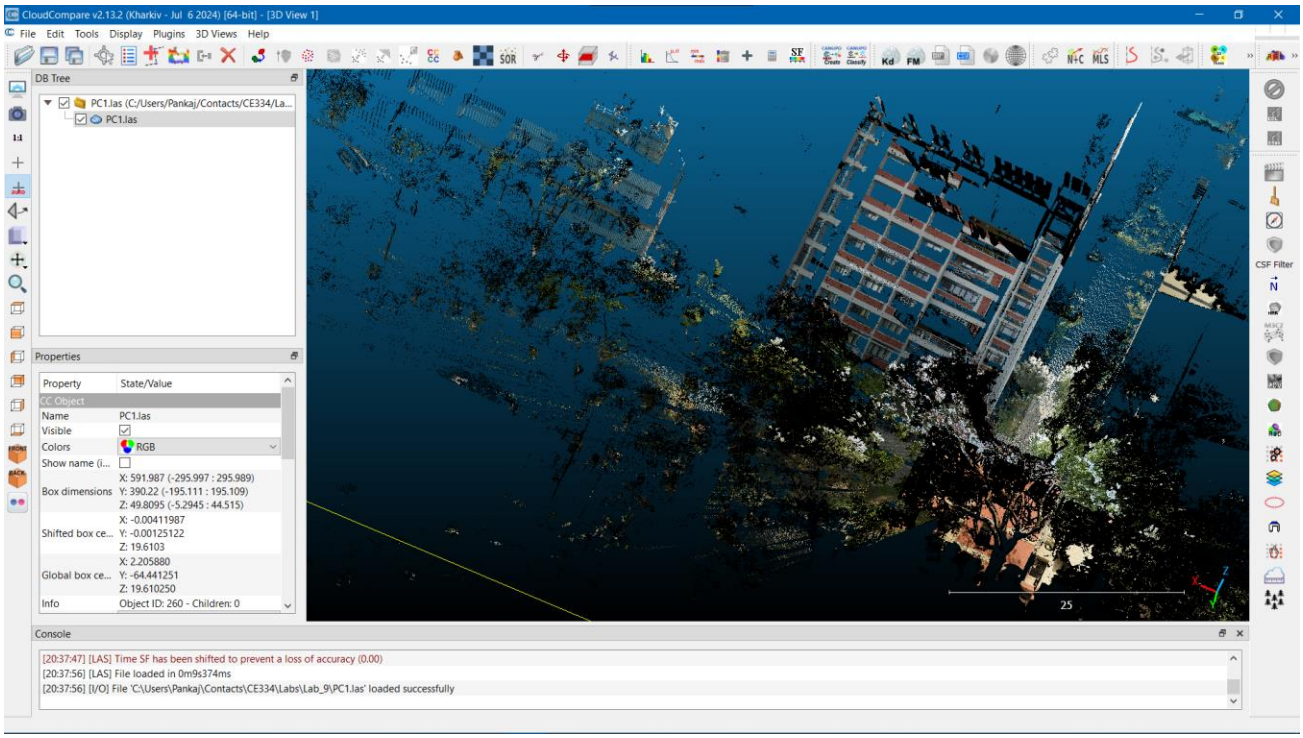
8.4 Export

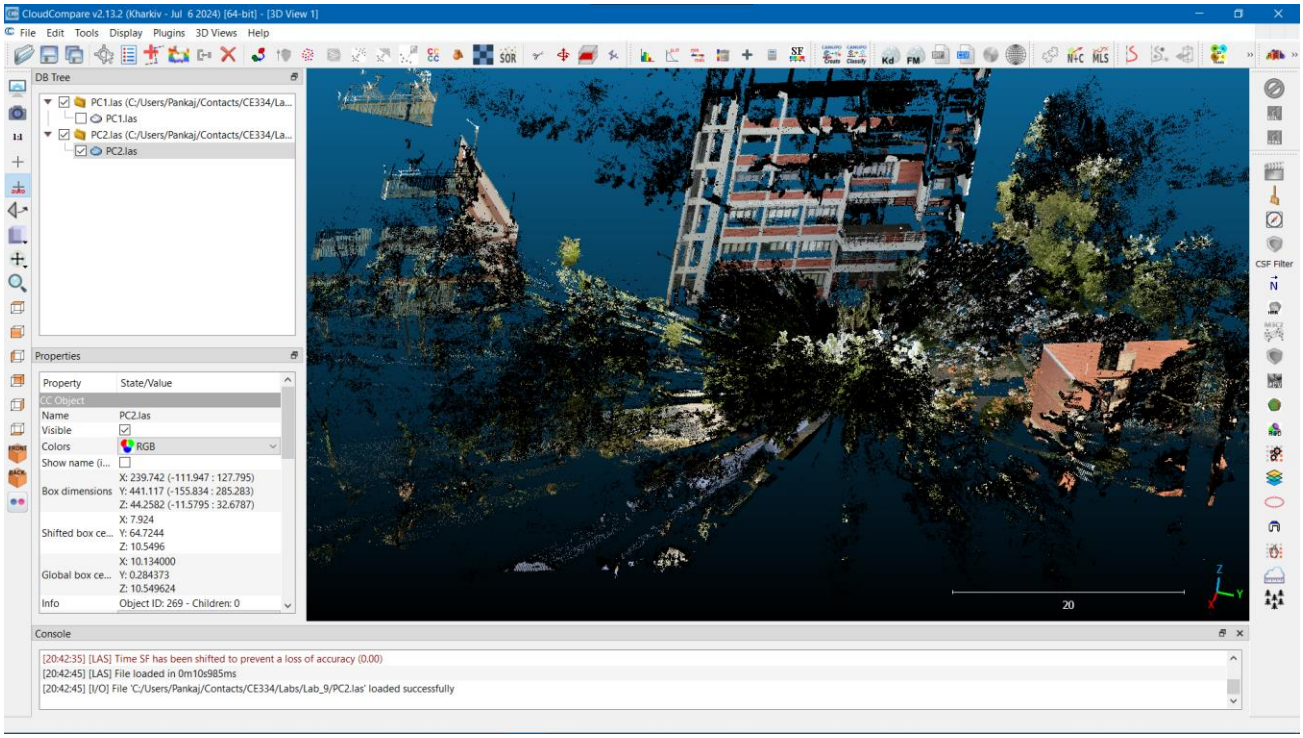
- The preprocessed and registered point cloud was exported in .las format for further analysis.

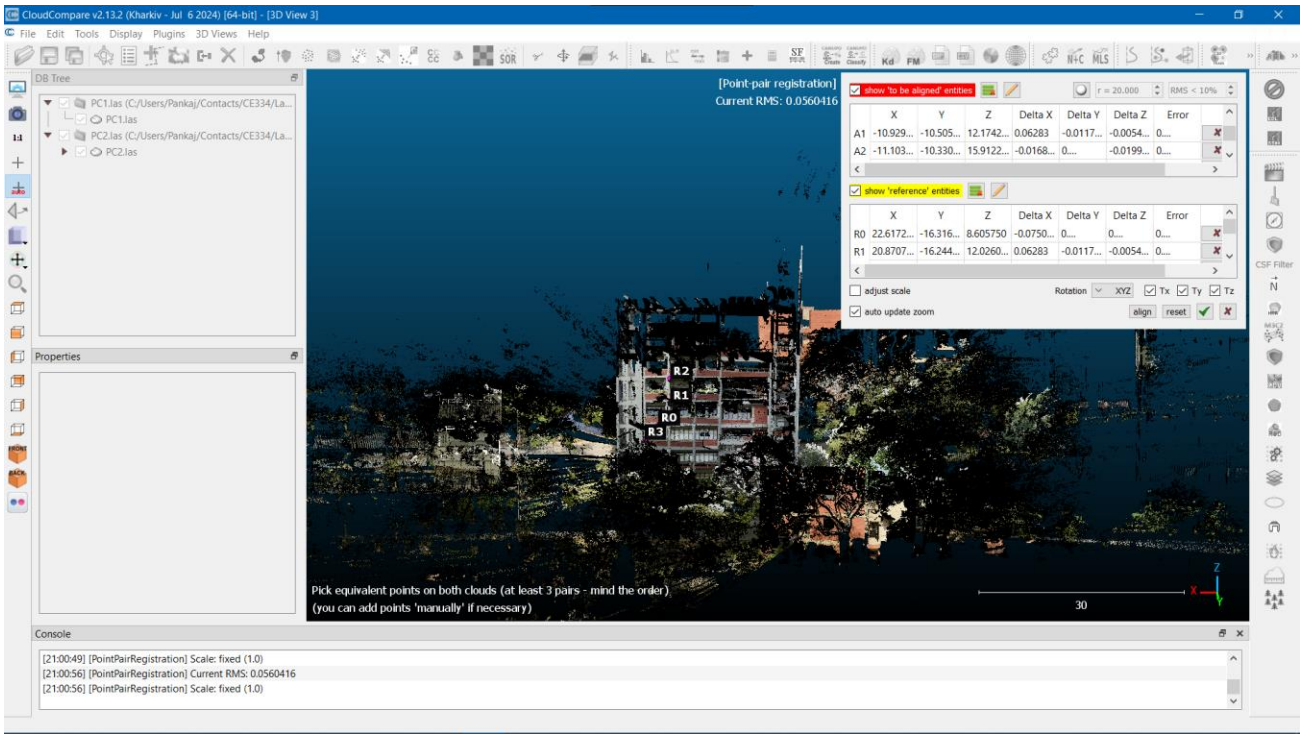
9. Analysis of Scan Quality

9.1 Data Density Variation

Point density varies across the scan depending on the distance from the scanner, the angle of incidence, and the resolution settings used. Regions closer to the scanner exhibit higher point density, while distant or obliquely-scanned surfaces have sparser coverage. The distribution of point density across the registered point cloud is described below:


CloudCompare v2.13.2 (Kharkiv - Jul 6 2024) [64-bit] - [3D View 1]
File Edit Tools Display Plugins 3D Views Help
DB Tree
PC1.las (C:/Users/Pankaj/Contacts/CE334/La...
PC1.las
Properties
Property State/Value
CC Object
Name PC1.las
Visible
Colors RGB
Show name (i...
Box dimensions X: 591.987 (-295.997 : 295.989) Y: 390.22 (-195.111 : 195.109) Z: 49.8095 (-5.2945 : 44.515)
Shifted box ce... X: -0.00411987 Y: -0.00125122 Z: 19.6103
Global box ce... X: 2.205880 Y: -64.441251 Z: 19.610250
Info Object ID: 260 - Children: 0
Console
[20:37:47] [LAS] Time SF has been shifted to prevent a loss of accuracy (0.00)
[20:37:56] [LAS] File loaded in 0m9s374ms
[20:37:56] [I/O] File 'C:\Users\Pankaj\Contacts\CE334\Labs\Lab_9\PC1.las' loaded successfully
25


CloudCompare v2.13.2 (Kharkiv - Jul 6 2024) [64-bit] - [3D View 1]
File Edit Tools Display Plugins 3D Views Help
DB Tree
PC1.las (C:/Users/Pankaj/Contacts/CE334/La...
PC1.las
PC2.las (C:/Users/Pankaj/Contacts/CE334/La...
PC2.las
Properties
Property State/Value
CC Object
Name PC2.las
Visible
Colors RGB
Show name (i...
Box dimensions X: 239.742 (-111.947 : 127.795) Y: 441.117 (-155.834 : 285.283) Z: 44.2582 (-11.5795 : 32.6787)
Shifted box ce... X: 7.924 Y: 64.7244 Z: 10.5496
Global box ce... X: 10.134000 Y: 0.284373 Z: 10.549624
Info Object ID: 269 - Children: 0
Console
[20:42:35] [LAS] Time SF has been shifted to prevent a loss of accuracy (0.00)
[20:42:45] [LAS] File loaded in 0m10s985ms
[20:42:45] [I/O] File 'C:/Users/Pankaj/Contacts/CE334/Labs/Lab_9/PC2.las' loaded successfully
20



9.2 Intensity Information Variation

The intensity (reflectance) values in the point cloud reflect the material properties and surface geometry of scanned objects. Highly reflective surfaces such as registration targets and painted walls return higher intensity values, while absorptive or wet surfaces show lower intensity. The following observations were made were not calculated due to lack of computational power.

9.3 Shadowing and Occlusion Effects

Shadowing effects occur when parts of the scene are occluded from the scanner's line of sight. These result in data gaps or 'shadow zones' in the point cloud.

Multiple scan positions partially addressed these gaps.

10. Results and Discussion

10.1 Best Scanning Positions and Angles

The selected scan positions provided complementary perspectives of the object, resulting in comprehensive spatial coverage. Scan Position 1 was placed at a moderate distance with a near-frontal orientation, allowing clear capture of primary surfaces with minimal occlusion. Scan Position 2 was positioned at an offset angle, which helped capture hidden or shadowed regions that were not visible from the first position.

The combination of these positions improved overall data completeness by reducing blind spots caused by complex object geometry. Horizontal coverage was maximized due to the 360° field of view, while the vertical field of view ensured sufficient capture of elevation variations.

However, some limitations remained. Areas with sharp recesses or undercuts were still partially occluded due to line-of-sight constraints inherent to terrestrial LiDAR. Additionally, very oblique angles at Position 2 introduced slight reductions in point density and accuracy on certain surfaces.

Overall, the chosen scan positions were effective in balancing coverage and efficiency, demonstrating the importance of multi-angle acquisition for complex geometries.

10.2 Impact of Scanner Settings on Data Quality

[The selected scanner settings had a significant influence on the quality and usability of the collected data. A fine angular resolution (0.04°) resulted in high point density, enabling detailed representation of surface features and improving the accuracy of subsequent analysis.

The scan speed (approximately 300,000 points per second) provided a balance between acquisition time and data quality. While higher scan speeds can reduce field time, they may introduce noise or reduce measurement precision. In this case, the chosen speed maintained acceptable accuracy while ensuring efficient data collection.

The wide field of view (360° horizontal and ~100° vertical) allowed extensive area coverage in a single scan, minimizing the number of required scan positions. However, this also increased the total number of collected points, leading to larger file sizes and longer processing times.

A key trade-off observed was between **resolution and data volume**. Higher resolution improved detail but significantly increased file size and processing complexity. Similarly, wider coverage reduced the need for multiple scans but introduced redundant data in overlapping regions.

Overall, the selected settings achieved a good compromise between detail, efficiency, and manageability of the dataset.

10.3 Effectiveness of Target Placements

The target placement strategy was generally effective in supporting accurate registration between multiple scan positions. Targets were distributed across the scene with sufficient spatial separation and visibility from both scan positions, ensuring robust alignment.

The registration process yielded low RMS residuals (typically within acceptable limits for terrestrial LiDAR surveys), indicating good consistency between scans. Most targets were clearly detected and contributed positively to the transformation accuracy.

However, a few challenges were observed. Some targets placed at extreme angles or partially occluded locations resulted in weaker returns or reduced detection reliability. Targets positioned too close together also reduced geometric strength during registration.

From this analysis, several key lessons were identified:

- Targets should be **well distributed in 3D space** (not in a single plane).
- Ensure **clear line-of-sight** from multiple scan positions.
- Avoid placing targets at **sharp angles or near edges**.
- Maintain adequate spacing to improve registration stability.

In conclusion, the target configuration was largely successful, with minor improvements possible in placement strategy to further enhance registration accuracy.

11. Recommendations for Future Scanning

- Increase the number of scan positions to reduce shadow zones behind large occluding objects.

- Place registration targets at three or more distinct elevation levels to improve registration geometry and reduce planimetric errors.
- Use higher resolution scans near objects of interest while employing coarser resolution for background regions to optimise scan time.
- Conduct a pre-visit to prepare a detailed scan plan, including estimated scan times and target layout map.
- Ensure that the camera is calibrated before mounting on the scanner for accurate point cloud colourization.

12. Conclusion

This laboratory exercise provided practical exposure to terrestrial LiDAR data collection using the Riegl TLS scanner. The scanning campaign successfully captured three-dimensional point cloud data of [describe the study area/object] from [number] scan positions. Post-processing in RiScan Pro enabled scan registration with an RMS error of [value], colourization using the mounted camera, and export in .las format for further analysis. Key observations regarding data density, intensity variation, and occlusion effects were documented and discussed. The exercise reinforced the importance of strategic scanner positioning, careful target placement, and appropriate parameter selection in achieving high-quality TLS data.